

Decent Coaching Classes
X SB
Academic Conquest - VI — Answer Key

Q.1 (A)

- (1) (B) 9% [CGST = SGST = half of 18%]
(2) (C) $1/2$ [prime numbers on a die : 2, 3, 5 $\rightarrow 3/6 = 1/2$]
(3) (B) (5, 12, 13) [$5^2 + 12^2 = 25 + 144 = 169 = 13^2$]
(4) (C) $\sec^2\theta$

Q.1 (B)

- (1) $a = 2, b = -5, c = 2. \quad b^2 - 4ac = 25 - 16 = 9$
(2) $a = 3, d = 4. \quad t_{10} = a + 9d = 3 + 36 = 39$
(3) $d(A, B) = \sqrt{[(4-2)^2 + (1-3)^2]} = \sqrt{(4+4)} = \sqrt{8} = 2\sqrt{2}$
(4) Ratio of areas = $(3/5)^2 = 9 : 25$

Q.2 (A) (any two)

- (1) Adding (I) and (II) : $3x = 9 \therefore x = 3$
Putting $x = 3$ in (I) : $2 \times 3 + y = 7 \therefore y = 1$
 $\therefore$ Solution is $(x, y) = (3, 1)$
(2) $QS^2 = PS \times SR = 9 \times 4 = 36 \therefore QS = 6$
(3) $\sec^2\theta = 1 + \tan^2\theta = 1 + (3/4)^2 = 1 + 9/16 = 25/16$
 $\therefore \sec\theta = 5/4, \cos\theta = 1/\sec\theta = 4/5$

Q.2 (B) (any two)

- (1) Putting $x = 3 : k(3)^2 - 10(3) + 3 = 0 \Rightarrow 9k - 30 + 3 = 0 \Rightarrow 9k = 27 \therefore k = 3$
(2) GST = 18% of 586 = $18/100 \times 586 = ₹ 105.48$ (CGST = SGST = ₹ 52.74)
Price for the customer = $586 + 105.48 = ₹ 691.48$
(3) Midpoint = $((22 + 0)/2, (20 + 16)/2) = (11, 18)$

Q.3 (A) (any one)

- (1) $n(S) = 4; A = \{ HH, HT, TH \}; n(A) = 3$
 $P(A) = n(A)/n(S) = 3/4$
(2) $81/225 = 144/x^2 \therefore 9/15 = 144/x^2 \therefore (3/5) = 12/x$ (taking square roots)
 $\therefore x = 12 \times 5/3 = 20 \text{ cm}$

Q.3 (B) (any two)

- (1) Three-digit numbers divisible by 5 : 100, 105, 110, ..., 995 (an A.P.)
 $a = 100, d = 5, tn = 995$
 $tn = a + (n-1)d \Rightarrow 995 = 100 + (n-1) \times 5 \Rightarrow (n-1) \times 5 = 895 \Rightarrow n-1 = 179 \Rightarrow n = 180$
 $\therefore$ There are 180 such numbers.
(2) Class marks (xi) and products (fi xi) :

Class	Class mark (xi)	Frequency (fi)	fi $\times$ xi
1 – 2	1.5	17	25.5
2 – 3	2.5	13	32.5
3 – 4	3.5	10	35
4 – 5	4.5	7	31.5

5 – 6	5.5	3	16.5
Total		$\Sigma fi = 50$	$\Sigma fi xi = 141$

$$\text{Mean} = \Sigma fi xi / \Sigma fi = 141 / 50 = 2.82 \text{ litres}$$

$$\begin{aligned} (3) \text{ LHS} &= (\sec\theta + \tan\theta)(1 - \sin\theta) \\ &= (1/\cos\theta + \sin\theta/\cos\theta)(1 - \sin\theta) \\ &= [(1 + \sin\theta) / \cos\theta] \times (1 - \sin\theta) \\ &= (1 - \sin^2\theta) / \cos\theta = \cos^2\theta / \cos\theta = \cos\theta = \text{RHS} \end{aligned}$$

Q.4 (any one)

- (1) Let the father's age be x years and the son's age be y years.

$$x + 2y = 70 \dots (I) \quad \text{and} \quad 2x + y = 95 \dots (II)$$

$$\text{From (I) : } x = 70 - 2y. \text{ Putting in (II) : } 2(70 - 2y) + y = 95 \Rightarrow 140 - 3y = 95 \Rightarrow y = 15$$

$$\therefore x = 70 - 30 = 40$$

$$\therefore \text{Father's age} = 40 \text{ years ; son's age} = 15 \text{ years.}$$

- (2) Let the distance of the ship from the lighthouse be d metres.

$$\text{Angle of elevation of the top from the ship} = \text{angle of depression} = 60^\circ$$

$$\tan 60^\circ = 90 / d \Rightarrow \sqrt{3} = 90 / d \Rightarrow d = 90 / \sqrt{3} = 30\sqrt{3}$$

$$d = 30 \times 1.73 = 51.9 \text{ m}$$

$$\therefore \text{The ship is } 51.9 \text{ m from the lighthouse.}$$

- (3) (i) Total investment = number of shares $\times$ MV = $50 \times 80 = ₹ 4000$

$$(ii) \text{ Dividend per share} = 20\% \text{ of FV} = 20/100 \times 100 = ₹ 20$$

$$\text{Total dividend} = 50 \times 20 = ₹ 1000$$

$$(iii) \text{ Rate of return} = (\text{total dividend} / \text{total investment}) \times 100 = 1000/4000 \times 100 = 25\%$$